Facial recognition of people with masks using MediaPipe Facemesh and deep learning algorithms

Mansour Hesabi Moghaddam[1], hamid reza ghaffary[2], Mahdi khazaiepoor[1]

[1]Computer Engineering Department, Birjand Branch, Islamic Azad University, Birjand, Iran

[2]Computer Engineering Department, Birjand Branch, Islamic Azad University, Ferdos Branch, Khorasan, Iran

Abstract

In response to the challenges facial recognition technologies face due to widespread mask usage during health crises like COVID-19, we introduce an innovative solution combining MediaPipe Facemesh with advanced deep learning to enhance recognition accuracy for individuals wearing masks. Our method employs a dual-stage deep learning framework, integrating with hybrid meta-heuristic algorithms, to achieve high precision in masked face recognition across various applications. A key innovation is the use of parallel algorithmic strategies for distinguishing masked from unmasked faces, alongside the introduction of a novel convolutional neural network (CNN) design optimized for efficient data clustering and feature extraction from masked images. This approach not only boosts accuracy but also scalability, adapting to increasing data volumes through two advanced combined algorithms. Our research marks a significant advancement in masked face recognition technology, offering improved accuracy and efficiency with significant implications for security and public health. This streamlined method represents a promising step towards addressing the limitations of mask usage in facial recognition, setting the stage for future technological advancements in the field.

Keywords: Facial Recognition, Masks, MediaPipe Facemesh, Deep Learning, Hybrid Meta-Heuristic Algorithms, Convolutional Neural Networks (CNN), Data Clustering, Feature Extraction.

Introduction

The Covid-19 pandemic has elevated the importance of facial recognition technology for individuals wearing masks, presenting a critical research area within artificial intelligence (AI) and image processing. Utilizing MediaPipe Facemesh for feature extraction and deep learning algorithms for model training, researchers aim to maintain security, healthcare, and access control system efficacy despite facial obstructions caused by masks. This technology's relevance extends across various applications, highlighting its potential in combating virus spread and enhancing public and digital security [1-10].

However, the adaptation of facial recognition to accommodate masked faces introduces significant challenges. The presence of masks alters facial features, complicating accurate recognition. Environmental factors like lighting and image quality further affect performance. Ongoing research seeks to refine algorithms and improve model accuracy, addressing these obstacles [11-20].

Efforts in AI and image processing focus on overcoming the diminished accuracy of masked facial recognition through advanced neural networks, integrating multiple data inputs, and developing new algorithms. Future technologies, including RFID and QR codes, offer alternative identification methods in scenarios where facial recognition may be impractical. Additionally, enhancing device precision and adopting 3D sensing could further improve recognition rates [8-18].

This paper proposes a novel approach utilizing a parallel two-stage deep learning method combined with hybrid meta-heuristic algorithms to increase detection accuracy for individuals with masks. By clustering data and employing parallel processing, the approach aims to reduce redundancy and increase efficiency. This method's contributions include an innovative algorithmic framework, integration of advanced technologies, and the potential for wide application across security, public health, and e-commerce sectors. The advancements proposed here signify a substantial step forward in masked facial recognition technology, promising to address current limitations and expand its utility in the face of ongoing health crises. The contributions of this manuscript are as follows:

The manuscript makes several significant contributions to the field of masked face recognition:

The manuscript presents several groundbreaking contributions to the domain of masked face recognition, each designed to advance the current state-of-the-art and address the unique challenges presented by the widespread use of facial masks. These contributions include:

1. **Innovative Parallel Algorithmic Framework**: Introducing a cutting-edge approach that employs parallel algorithms specifically for the recognition of masked faces, this framework not only elevates the precision of detection but also offers a scalable solution adept at managing extensive datasets efficiently.
2. **Advanced Technological Integration**: The research pioneers the integration of 3D sensing and high-definition imaging technologies to refine the accuracy of masked face

recognition systems. These advancements bolster the system's capability to discern and analyze facial features shrouded by masks, enhancing overall detection performance.

3. **Hybrid Meta-Heuristic Algorithms**: This manuscript introduces hybrid meta-heuristic algorithms as a novel contribution to facial recognition technology. By amalgamating the strengths of various algorithms, this approach yields superior accuracy and robustness in the identification of individuals wearing masks, marking a significant step forward in the field.

Proposed method

In this section, we present the proposed method for improving masked face recognition using a parallel two-stage deep learning approach along with hybrid meta-heuristic algorithms. The method involves data clustering, feature selection, outlier data removal, and a consensus-based approach to achieve higher accuracy in masked face recognition. Figure 1 provides an overview of the proposed method based on deep learning, and Figure 2 illustrates the use of a correlation matrix in the process.

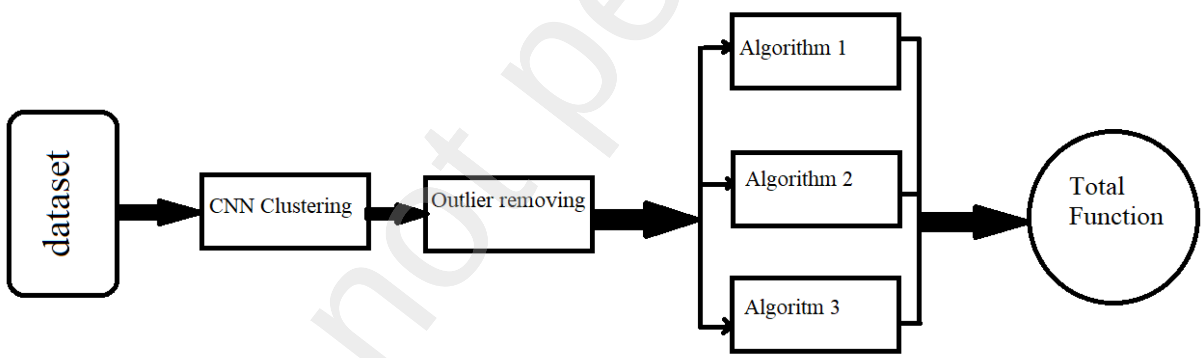

Figure 1: Proposed Method Based on Deep Learning

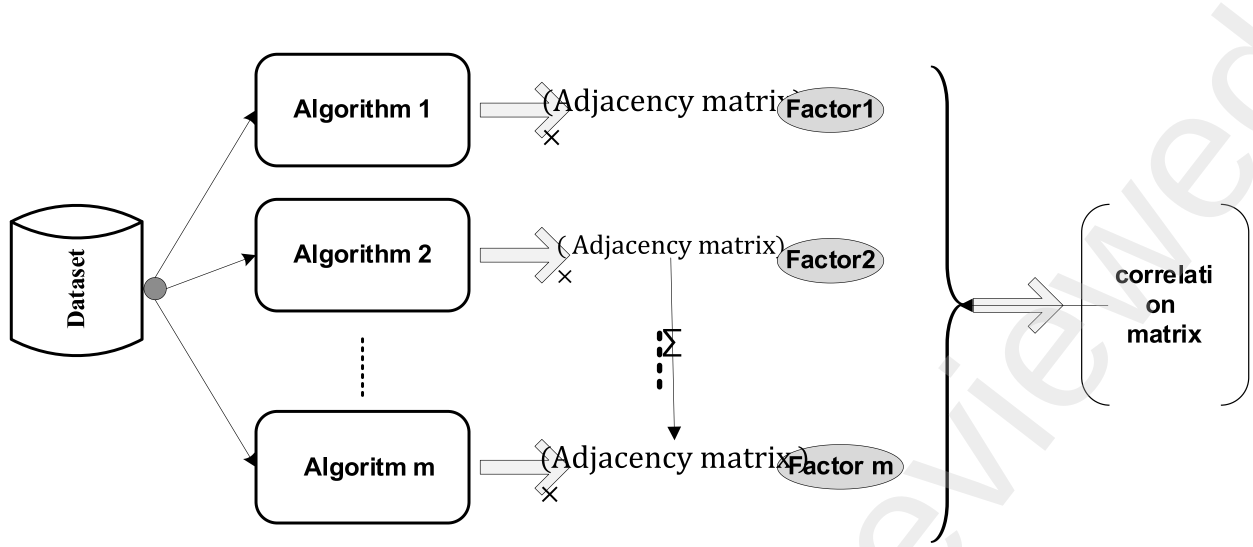

Figure 2: Proposed Method with Correlation Matrix

Clustering with Convolutional Neural Network (CNN)

In the realm of facial recognition technology, the advent of widespread mask usage poses unprecedented challenges. These challenges necessitate innovative approaches to ensure the accuracy and reliability of facial recognition systems. A cornerstone of such innovative methodologies is the application of Convolutional Neural Networks (CNNs) in conjunction with data clustering. This paper delves into the intricacies of this approach, highlighting the pivotal role of data segmentation in the overall process of masked face recognition.

Data Segmentation: A Critical Precursor to Effective Recognition

Data segmentation stands as a fundamental step in the masked face recognition system. Its significance lies in its direct influence on the subsequent stages, including feature extraction, algorithmic processing, and, ultimately, the accuracy of the facial recognition process.

Segmentation Criteria: Volume and Characteristics of Data

The segmentation of data in the context of masked face recognition is primarily governed by two key criteria: the volume of data and its characteristics.

1. **Volume of Data**
   - **Importance**: The volume of data, encompassing both the quantity of images and their resolution, plays a crucial role in determining the structure and size of the data clusters.
   - **Implementation**: Datasets with higher-resolution images or a larger number of images typically require more robust or numerous clusters to manage the increased computational complexity. In contrast, datasets with fewer or lower-resolution images can be effectively managed with fewer or simpler clusters.

2. **Characteristics of Data**
   - **Variations in Mask Types**: Different mask types, such as surgical, N95, and cloth masks, present unique challenges in feature extraction. The segmentation process, therefore, categorizes images based on the type of mask to tailor the feature extraction and recognition algorithms more effectively.
   - **Facial Features Visible Through Masks**: Identifying visible parts of the face, such as eyes, the nose bridge, and the forehead, is critical. The segmentation process groups images based on the visibility of these key facial features.
   - **Diversity in Background Conditions**: Factors like lighting, whether the setting is indoors or outdoors, and the presence of other objects or people, significantly affect recognition accuracy. The segmentation aims to group images with similar background characteristics, thus enhancing accuracy under specific conditions.

Execution of Data Segmentation

Data segmentation in this context is executed through a two-pronged approach:

- **Automated Segmentation**: Leveraging machine learning algorithms, such as K-means or DBSCAN, the system automatically segments the data based on predefined criteria. These algorithms analyze the image characteristics and sort them into clusters with similar features.
- **Manual Review and Adjustment**: Following the initial automated segmentation, a manual review is essential to ensure that the data has been accurately categorized. This step is vital for rectifying any anomalies and refining the segmentation process.

Impact on Subsequent Processes

The efficacy of data segmentation has a profound impact on the overall efficiency and effectiveness of the facial recognition system. By ensuring that data is correctly segmented, each subsequent algorithm or process operates on the most relevant subset of data. This targeted approach not only maximizes the accuracy of the facial recognition process but also enhances computational efficiency.

In conclusion, the integration of CNNs with a strategic data segmentation approach forms the bedrock of adapting facial recognition technology to the challenges posed by mask usage. This methodological advancement not only addresses current limitations but also sets the stage for future developments in the field of artificial intelligence and image processing.

Cluster Formation:

A critical component in the enhanced framework of masked face recognition using Convolutional Neural Networks (CNNs) is the strategic formation of data clusters. This section will explore the purpose and methodology of cluster formation, emphasizing its role in refining the facial recognition process under the unique constraints posed by mask usage.

Purpose of Cluster Formation

The primary objective of forming clusters within this context is to encapsulate and address specific challenges associated with masked face recognition. This focused approach allows for the tailored handling of different variables that masks introduce into the facial recognition process.

- **Targeted Approach**: By segregating the data into distinct clusters, the system can apply specialized algorithms and feature extraction methods that are optimally designed for each unique scenario.
- **Enhanced Accuracy**: This segmentation into clusters enables the system to process each subset of data with greater precision, thereby improving the overall accuracy of the facial recognition system.

Examples of Cluster Differentiation

To illustrate the application of cluster formation, consider the following examples:

1. **Cluster Focused on Surgical Masks**
   - **Characteristics**: This cluster includes images of faces covered with surgical masks. These masks typically have a uniform color and cover a significant portion of the lower face.
   - **Specialized Processing**: The algorithms and feature extraction methods in this cluster are fine-tuned to recognize the upper facial features, such as the eyes and forehead, which are less obscured by surgical masks. The uniformity in mask type allows for a more standardized approach to feature extraction within this cluster.
2. **Cluster Concentrating on Cloth Masks with Varying Patterns**
   - **Characteristics**: This cluster is dedicated to faces with cloth masks, which often feature diverse colors, patterns, and textures.
   - **Adapted Methodology**: Given the variability in mask appearances, the algorithms here are adapted to handle a wider range of visual inputs. The system might employ advanced techniques to differentiate between the mask's fabric and the wearer's skin, ensuring accurate identification despite the visual complexity of the masks.

The aim of cluster formation is to segregate the dataset into distinct groups, each tailored to specific challenges in masked face recognition. This can be formalized as:

Let $D = \{d_1,d_2,\ldots,d_n\}$ be the dataset where each d_i represents an image. The goal is to partition D into k clusters $C = \{C_1,C_2,\ldots,C_k\}$ such that each cluster C_j is optimized for a specific type of mask challenge.

Implementation of Cluster Formation

The implementation of cluster formation in masked face recognition involves several key steps:

- **Data Categorization**: Initially, the system categorizes the incoming data based on predefined characteristics, such as mask type, visible facial features, and background conditions.
- **Algorithmic Alignment**: For each cluster, specific algorithms are aligned to best address the challenges unique to that subset of data. This includes selecting the most effective convolutional neural network architectures and feature extraction techniques suitable for each cluster.
- **Continuous Refinement**: As the system processes more data, it continually refines the cluster formation based on feedback and performance metrics. This iterative process ensures that the clusters remain optimally aligned with the evolving challenges in masked face recognition.

The process of forming clusters can be mathematically represented using clustering algorithms. For instance, using the K-means algorithm, the objective is to minimize the within-cluster sum of squares (WCSS):

$$\text{Minimize}\, WCSS(C) = \sum_{j=1}^{k} \sum_{d_i \in C_j} \| d_i - \mu_j \|^2$$

where μ_j is the mean of points in C_j. The clusters are continuously refined based on performance metrics. Let P(Cj) be the performance metric for cluster Cj, the refinement can be mathematically modeled as:

$$C_j^{(\text{new})} = \text{Refine}\left(C_j, P(C_j)\right)$$

where $C_j^{(\text{new})}$ is the updated cluster after refinement.

CNN-Based Clustering and Classification

In the context of masked face recognition, CNN-based clustering and classification play a pivotal role. This section delves into the architecture and optimization of CNNs, providing a comprehensive overview with detailed formulas and relationships.

1. CNN Architecture

The architecture of a Convolutional Neural Network (CNN) for this application is custom-designed to efficiently group similar data points, especially those altered by mask-wearing. The architecture typically consists of the following layers:

- **Convolutional Layers (ConvLayer)**
    - Design: Convolutional layers apply a number of filters to the input data to create feature maps.
    - Mathematical Representation: Let X be the input data, F be the filter, and b be the bias. The output O of a convolutional layer is given by $O=F*X+b$, where "$*$" denotes the convolution operation.
- **Pooling Layers (PoolingLayer)**
    - Function: Pooling layers reduce the spatial dimensions (width and height) of the input data, thereby reducing the number of parameters and computation in the network.
    - Common Types: Max Pooling and Average Pooling.
- **Fully Connected Layers**
    - Design: These layers connect every neuron in one layer to every neuron in the next layer, aiding in the classification task.
    - Mathematical Expression: Given an input vector V, the output Y is calculated as $Y=W\cdot V+b$, where W represents the weight matrix and bb the bias vector.

The general architecture can be expressed as:

$$\text{CNNarchitecture} = \text{ConvLayer } 1 \rightarrow \text{PoolingLayer} \rightarrow \text{ConvLayer } 2 \rightarrow \text{FullyConnectedLayer}$$

1. **Optimization**:

Optimizing a CNN involves refining its ability to discern subtle nuances in data, especially under varying mask-wearing conditions.

- **Objective**
    - Enhance the CNN's capacity to recognize and differentiate masked faces accurately.
- **Techniques**
    - **Dropout**
        - Reduces overfitting by randomly omitting units from the neural network during training.
        - Formula: $D(vi)=vi*N(1,p)$, where vi is a neuron's output, and N(1,p) is a binary random variable determining if the neuron is dropped.
    - **Batch Normalization**
        - Stabilizes and speeds up learning by normalizing the input layer by adjusting and scaling the activations.

- Formula: $BN(x) = \gamma\left(\frac{x-\mu}{\sqrt{\sigma^2+\epsilon}}\right) + \beta$, where μ and σ^2 are the mean and variance of the batch, ϵ is a small constant, and γ,β are parameters to learn.

- **Hyperparameter Tuning**

Involves adjusting parameters like learning rate, number of epochs, and number of layers to find the most efficient network configuration.

Feature Engineering

In the intricate process of masked face recognition, feature engineering plays a crucial role, especially in the context of a Convolutional Neural Network (CNN) framework. The essence of feature engineering lies in two fundamental components: feature extraction and feature selection.

The first component, feature extraction, involves the CNN systematically identifying and extracting vital features from facial data obscured by masks. This extraction is not merely a process of selecting data points but rather an art of discerning and highlighting the most informative aspects of the facial data. In the scenario of masked faces, the CNN's focus shifts predominantly to those facial features less affected by masks. These key features typically include the shape and position of the eyes, the area of the nose bridge, and the forehead. The strategic emphasis on these areas allows the CNN to capture essential facial characteristics that remain visible and unobstructed by masks, thereby maintaining the integrity and reliability of the facial recognition process.

Once these critical features are extracted, the next step, feature selection, comes into play. The selection process is not arbitrary but is governed by specific criteria, primarily focusing on the relevance and discriminative power of the features in the context of masked face recognition. This step ensures that only the most significant features that contribute to the accurate recognition of masked faces are retained, thereby optimizing the performance of the recognition system. To achieve this, sophisticated techniques like Principal Component Analysis (PCA) and t-Distributed Stochastic Neighbor Embedding (t-SNE) are employed. These methods are instrumental in reducing the dimensionality of the feature space without compromising the essential characteristics of the data.

PCA serves as a powerful tool for identifying and emphasizing variation and bringing out strong patterns in a dataset. By transforming the original data into a set of linearly uncorrelated orthogonal variables, PCA highlights the most significant features with the highest variances. On the other hand, t-SNE, a probabilistic technique, excels in the visualization of high-dimensional data by converting similarities between data points to joint probabilities. This method is particularly adept at creating a map of the original features in a reduced dimensional space while preserving the relative distances, thereby ensuring that the intrinsic properties of the data are maintained.

The integration of PCA and t-SNE in processing CNN-extracted features can be represented by the equation:

$$\text{SelectedFeatures} = \text{PCA}(\text{t} - \text{SNE}(\text{CNNextractedFeatures}))$$

This equation encapsulates the sequential application of t-SNE for initial dimensionality reduction followed by PCA for further refinement and selection of features. The outcome is a set of highly optimized, relevant, and discriminative features suitable for the effective recognition of individuals with masked faces.

Outlier Data Removal

The process of outlier data removal in the context of masked face recognition using CNNs is a critical step towards enhancing the accuracy and efficiency of the recognition system. This process involves two main stages: the identification of outliers and redundant features, and the subsequent steps for data cleaning.

Identification of Outliers and Redundant Features

The initial phase in outlier data removal is the identification of anomalous data points within each cluster. This is achieved using sophisticated anomaly detection techniques such as the Isolation Forest method and the One-Class Support Vector Machine (One-Class SVM).

- **Isolation Forest**: This technique works effectively on high-dimensional data and isolates anomalies instead of profiling normal data points. It operates by randomly selecting a feature and then randomly selecting a split value between the maximum and minimum values of the selected feature. Anomalous data points are those that have shorter path lengths in the forest, as they are easier to isolate.
- **One-Class SVM**: This method is particularly useful for datasets where the outliers are not well-represented. It learns a decision function for outlier detection, identifying data points that deviate from the norm. One-Class SVM works by defining a decision boundary that separates the majority of data points from the outliers.

Data Cleaning Steps

Once outliers and redundant features are identified, the next step is the data cleaning process, which is critical to maintain the integrity of the data used for training the CNN. This process involves:

1. **Detecting Outliers**: Outliers are detected using the aforementioned statistical methods. This involves analyzing the data to identify points that deviate significantly from the norm. The deviation is typically measured in terms of distance or similarity to the majority of data points in the dataset.
2. **Assessment and Removal**: After the identification of potential outliers, a careful assessment is undertaken. This assessment is crucial to differentiate between genuine outliers and unusual but valid data points. The aim is to remove genuine outliers that could

skew the results of the facial recognition process while preserving critical information that contributes to the diversity and richness of the dataset.

The removal of these outliers and redundant features enhances the overall system in several ways:

- **Improved Efficiency**: By eliminating irrelevant data, the computational load on the system is reduced, leading to faster processing times.
- **Increased Accuracy**: Removing outliers ensures that the CNN is not trained on anomalous data, which could lead to inaccurate recognition results.

Table 1 and figure 3 provide Summary CNN-Based Clustering Process and the proposed method for this research.

Table 1 Summary Table of CNN-Based Clustering Process

Step	Description	Techniques/Tools Used
Data Segmentation	Dividing data based on volume and characteristics	-
CNN Architecture	Designing layers for efficient data grouping	Convolutional layers, Pooling layers, Activation functions
Optimization	Refining the CNN for subtle feature detection	Dropout, Batch normalization, Hyperparameter tuning
Feature Extraction	Identifying pivotal masked face features	CNN
Feature Selection	Selecting relevant features	PCA, t-SNE
Outlier Data Removal	Eliminating redundant and outlier data	Isolation Forest, One-Class SVM

Figure 3 the details of proposed method

Parallel Algorithms

In the realm of facial recognition technology, especially in the context of masked faces, the use of parallel algorithms represents a significant advancement. This approach involves deploying multiple algorithms simultaneously, each tailored to recognize faces with masks based on specific characteristics identified in their respective clusters. The underlying principle is to enhance the accuracy and reliability of facial recognition in various environments, including those where individuals may cover their faces for various reasons.

The integration of multiple algorithms in this process is crucial. Each algorithm within the proposed method operates independently, analyzing faces with masks and removing outlier data through techniques such as modularity maximization and correlation matrix analysis. This independent operation allows each algorithm to focus on specific features and challenges associated with its cluster of masked faces, leading to more refined and accurate recognition.

Architectural Specifications of the CNN

The architecture of the Convolutional Neural Network (CNN) used in this research is intricately designed. It comprises multiple layers, including convolutional layers, pooling layers, and fully connected layers. The convolutional layers are responsible for extracting features from the input images, while the pooling layers reduce the dimensionality of these features, leading to more efficient processing. The fully connected layers, at the end of the architecture, play a pivotal role in the classification process. Activation functions such as ReLU or sigmoid are employed based on their performance in preliminary tests, ensuring the best possible outcome in terms of feature activation and non-linearity introduction.

Training and Optimization Protocols

The training of this CNN involves using a carefully curated dataset, representative of the varied scenarios encountered in masked face recognition. The training procedures include feeding the network with input data, forward propagation to calculate the output, and backward propagation to update the network's weights and biases. Optimization strategies such as stochastic gradient descent, Adam optimizer, or RMSprop are employed, depending on their effectiveness in reducing the loss function and improving the network's performance.

Performance Evaluation Metrics

To evaluate the performance of the CNN, metrics such as accuracy, precision, recall, and F1-score are used. Accuracy measures the proportion of correctly identified instances, precision evaluates the correctness of positive identifications, recall assesses the ability of the model to find all relevant cases, and the F1-score provides a balance between precision and recall.

Details of the First Algorithm: MediaPipe Facemesh

In the first algorithm, MediaPipe Facemesh, designed by Google, is utilized. This algorithm excels in detecting 68 key points on the human face, including areas around the nose, eyes, and mouth. Its capability extends to recognizing facial movements, evaluating eyeglasses, AR clothing, and identifying individuals in security footage.

For masked face recognition, the MediaPipe Facemesh algorithm is combined with other algorithms to enhance its efficacy. These include vein recognition methods for improved accuracy based on mask types and facial details, texture-based methods for extracting detailed information from masked faces, attention-based methods focusing on significant areas of a masked face, deep learning methods employing CNN and LSTM for sophisticated feature analysis, and image processing methods using Gabor filters for enhanced pattern recognition on the face.

Proposed Hybrid Method

Figure 4 shows the flowchart of first algorithm. The proposed hybrid method combines the strengths of CNN and LSTM (Long Short-Term Memory) networks. This combination ensures not only high accuracy in detection but also maintains reasonable processing time and computational cost. The process involves the following steps:

1. Receiving video or image input with a mask.
2. Pre-processing the input frames to enhance image quality.
3. Performing face recognition using the hybrid CNN.
4. Post face detection, determining 468 facial landmarks even with a mask, using advanced CNN techniques.
5. Calculating 3D coordinates for each of these facial points.
6. Finally, presenting the results which include detailed information about the face, such as its position in the image.

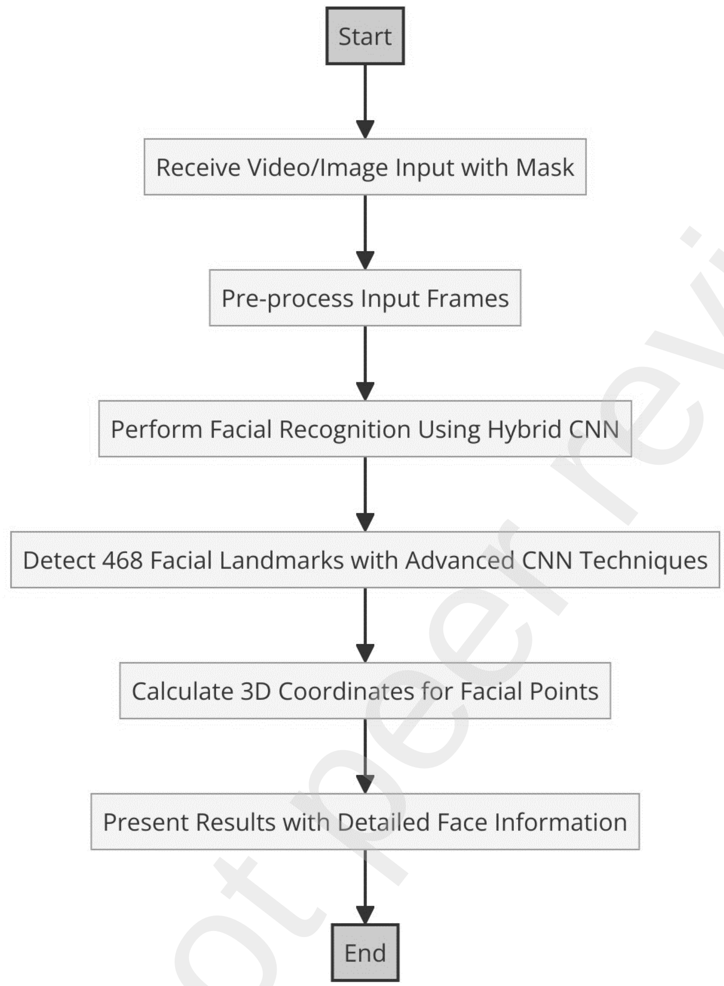

Figure 4 the flowchart of first algorithm

Details of the second algorithm

Figure 5 shows the flowchart of second algorithm. Another hybrid algorithm can be used for the unmasked data part. The second algorithm in the proposed method for facial recognition, particularly focusing on unmasked data, is a sophisticated hybrid algorithm that employs a combination of Information Gain (IG), Genetic Algorithms (GA), and Genetic Programming (GP). This algorithm is particularly designed to enhance the recognition process by selecting and classifying features effectively, thereby accommodating both masked and unmasked individuals.

Key Components of the Second Algorithm

1. **Feature Selection Using Information Gain (IG):**

- **Creating a Classification Feature:** This step involves identifying a feature that effectively classifies the data into distinct categories. It serves as a foundational step in the feature selection process.
- **Classification Entropy Calculation:** For each class in the dataset, the algorithm calculates the entropy. Entropy, in this context, measures the uncertainty or impurity in the class distribution. The lower the entropy, the higher the purity of the class.
- **Conditional Probability Calculation:** For each feature, the algorithm calculates the probability of all its possible values. This step is essential for understanding the distribution of each feature within the dataset.
- **Conditional Entropy Calculation:** Based on the probabilities obtained, the algorithm calculates the conditional entropy for each feature. This metric assesses the amount of information required to describe the outcome of a random variable, given the presence of another variable.
- **Information Gain Calculation:** The information gain for each feature is computed by subtracting its conditional entropy from the original entropy of the dataset. Information gain measures the effectiveness of a feature in classifying the dataset.
- **Feature Selection:** The features are then ranked according to their information gain, and the top 'k' features with the highest information gain are selected. This subset represents the most informative features for classification tasks.

2. **Feature Reduction Using Genetic Algorithms (GA):**
   - GA is applied to the selected features to further reduce their number. This reduction is crucial for eliminating redundancy and focusing on the most significant features for the classification process.
3. **Classification Using Genetic Programming (GP):**
   - Finally, GP is used to classify different types of masks in the dataset. GP evolves programs or models that can effectively distinguish between various mask types, thereby enhancing the facial recognition process.

The proposed method adapts to the increasing volume of data and the number of clusters. As the data grows and diversifies, necessitating more clusters, the first and second algorithms are used in alternation. This adaptive approach ensures that the system remains efficient and accurate, regardless of the scale and complexity of the dataset.

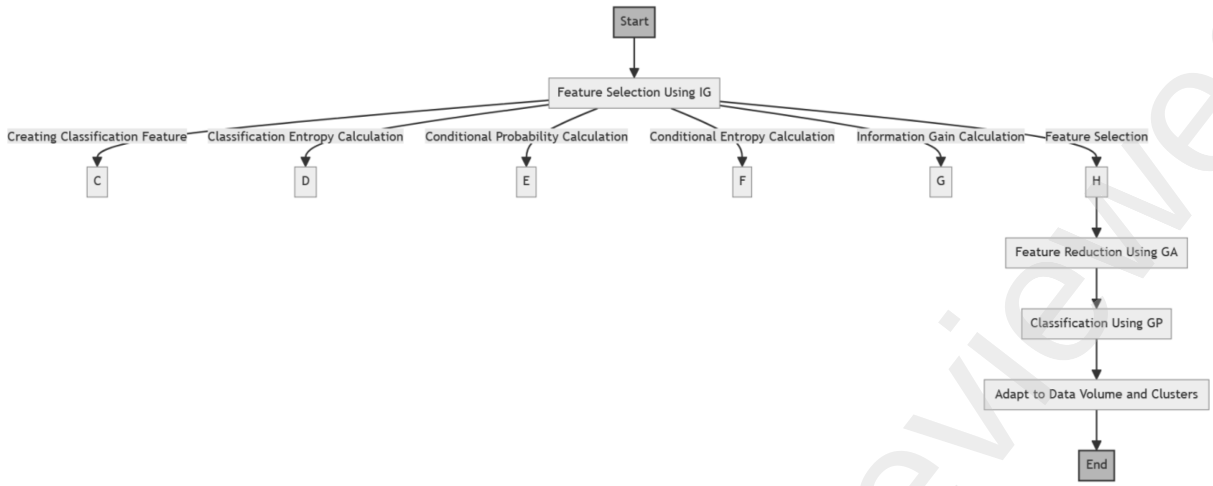

Figure 5 the flowchart of second algorithm

Proposed Deep Neural Network

In the proposed method presented above, a two-dimensional CNN and LSTM network is used for the neural network. Deep learning models are a group of models that can learn a hierarchy of features by constructing high-level features from low-level features, and in this way, feature extraction is automated. do These learning machines can be used in both supervised and unsupervised ways, and in both cases, they have shown competitive results in the areas of detection and signal processing. Convolutional neural networks are a class of deep models in which trainable filters and max pooling operators are applied one at a time on the input vectors, creating a hierarchy of features with increasing complexity. they become It has been shown that if these models are trained with specific settings, they can achieve advanced results in the fields of signal processing without relying on manual features. Multi-phase architectures and the integration of different features have also improved these results. The main core of convolutional networks is convolutional filters that act on the entire input vector. The final structure of the proposed method for the part of the neural network will be according to the flowchart in Figure 6. This structure includes the proposed two-dimensional method as well as added modules to improve performance. It should be noted that in the integration part, the neural network method is combined with the fuzzy method.

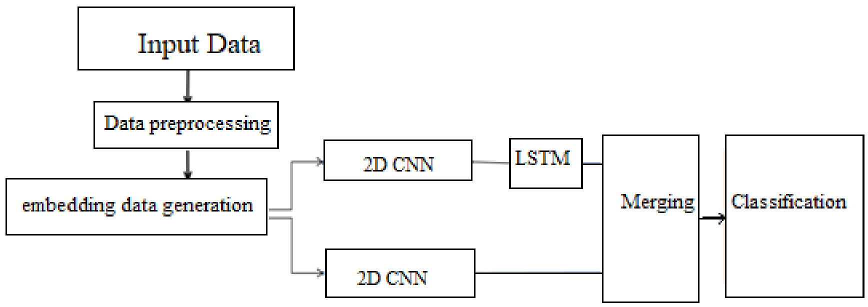

Figure 6 Flow chart of problem solving

In this proposed method, we prefer that the representation vectors are also part of the model learning process and we do not use ready generated vectors. With this method, we will ensure that the learned representation vectors will be suitable for the data used in this research because the existing trained vectors are for other specific applications, but our research focuses on microarray data.

Another innovation of our proposed method is that in addition to two-dimensional convolutional networks, LSTM networks have been used. And finally, the prediction made by both types of networks has been applied in the final decision. This is done in the integration layer. In our proposed solution, both of these outputs are used. According to the drawn flow diagram, the upper process of the diagram in the deep networks section uses the output one-dimensional vector from two-dimensional convolutional networks as a feature vector for training LSTM networks. Figure 7 shows the placement of representation vectors for training a two-dimensional convolutional network.

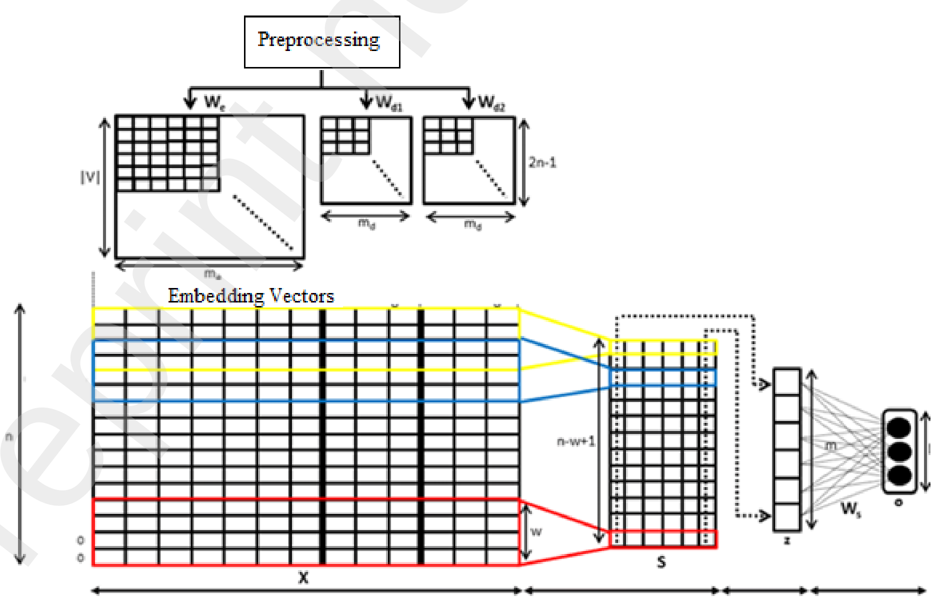

Figure 7 An example of the arrangement of representation vectors for convolutional network training

merge

According to the explanation of the proposed method as well as the explanations about the importance of using multi-flow structures and integrating different features in the previous section, in this research, a method to take advantage of different results and integrate them has been presented. . This type of integration is known as late integration because they combine the results in the final part of the model. Another type of merger is early mergers, which merge features with each other in the early stages of the process. In order to perform integration, the probabilities produced by the softmax layer of each network have been used. More precisely, each network is trained separately, and then when predicting the label of an expression, the probabilities generated by each network for that expression are first multiplied by the number and finally the maximum These new probabilities are considered as predictions made for the input expression. If we assume that P ⃗_2d (C|x) are the probabilities generated for an input expression by the proposed two-dimensional network and P ⃗_LSTM (C|x) are the probabilities generated for the same expression by the LSTM network, then P ⃗_new (C| x) The predicted points for that statement will be based on the following relationship. C is the set of labels of the dataset.

$$\vec{P}_{2d}(C = i \mid x,W,b) = \mathrm{softmax}(Wx + b) = \frac{e^{w_i x + b_i}}{\sum_j e^{w_j x + b_j}}$$

$$\vec{P}_{LSTM}(C = i \mid x,W,b) = \mathrm{softmax}(Wx + b) = \frac{e^{w_i x + b_i}}{\sum_j e^{w_j x + b_j}}$$

$$\vec{P}_{new}(C = i \mid x,W,b) = \vec{P}_{2d}(C = i \mid x,W,b) \otimes \vec{P}_{3d}(C = i \mid x,W,b)$$

$$\vec{P}_{2d} = [p_{2d}^1, p_{2d}^2, p_{2d}^3, \ldots, p_{2d}^l], \vec{P}_{3d} = [p_{3d}^1, p_{3d}^2, p_{3d}^3, \ldots, p_{3d}^l]$$

where ⊗ is the multiplication sign of the number of groups and l is equal to the number of groups. Finally, the predicted label for the target verbs will be the following relation (Chen and Dai 2021)

C_predict=argmax_i P(C=i │ x,W,b)

The advantage of this method is the ease of implementation. This ease comes from the fact that the two networks are trained separately and as a result there is no interference between the back propagation algorithms in the two networks. Figure 8 shows the integration layer. Figure 9 shows the flowchart of complete proposed method.

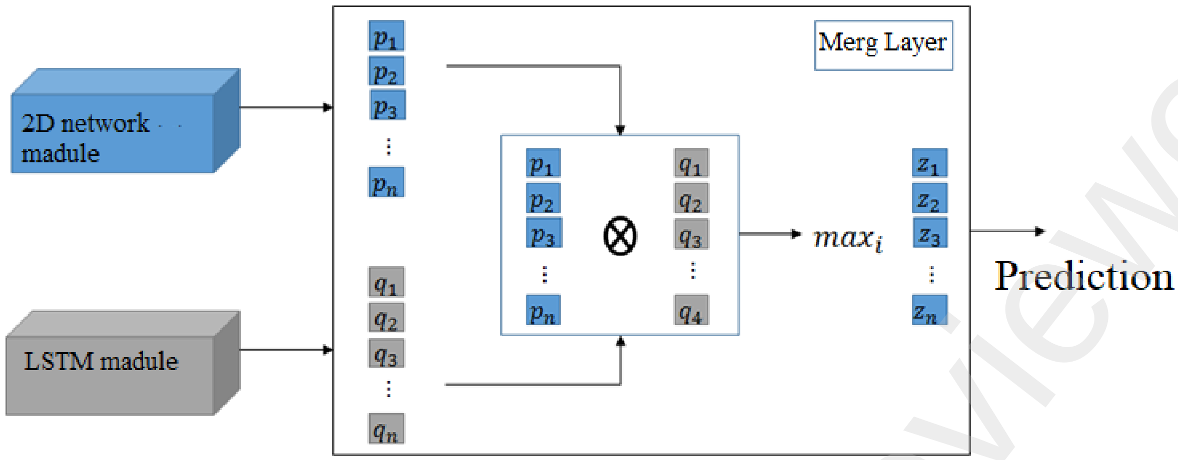

Figure 8 How the integration layer works to integrate the probabilities generated by the two-dimensional network and the LSTM network

To evaluate the classification results, three indicators, namely sensitivity, specificity and accuracy according to relation (1), (2) and (3) were used.

$$sensitivity = \frac{TP}{(TP + FN)} \quad (1)$$

$$specificity = \frac{TN}{(TN + FP)} \quad (2)$$

$$accuracy = \frac{(TP + TN)}{(TP + TN + FN + FP)} \quad (3)$$

True Positive Answer (TP): There are records in this category that are in the positive category and the classifier has correctly identified them as positive.

True Negative Answer (TN): There are records in this category that are in the negative category and the classifier has correctly identified them as negative.

False Positive Answer (FP): Records in this category that are in the negative category and have been incorrectly identified as positive by the classifier.

False Negative Answer (FN): Records in this category that are in the positive category and have been incorrectly identified as negative by the classifier.

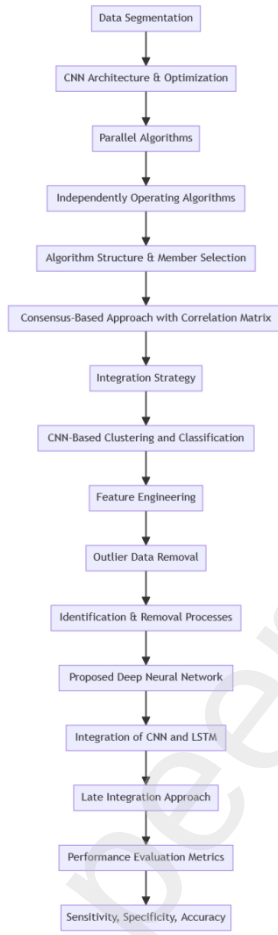

Figure 9 the flowchart of complete proposed method.

Dataset

The dataset employed in this research plays a pivotal role in developing and refining the facial recognition algorithms, particularly in the context of distinguishing between masked and unmasked faces. This dataset is robust and extensive, encompassing a wide range of scenarios and variations, which are essential for training and testing the algorithms effectively.

Characteristics of the Dataset:

1. **Diversity in Mask Wearing:** The dataset includes images of individuals wearing various types of masks, including surgical masks, cloth masks, N95 respirators, and other forms. This diversity is crucial for the algorithms to learn and adapt to different mask styles and how they affect facial features.
2. **Inclusion of Unmasked Individuals:** To ensure the system's efficacy in recognizing faces with and without masks, the dataset also contains images of individuals without any facial

coverings. This inclusion helps in refining the algorithm's ability to distinguish between masked and unmasked faces accurately.

3. **Volume of Data:** With over 1000 images, the dataset is relatively large, providing a comprehensive range of data necessary for effective training. The size of the dataset is significant for developing a robust model capable of handling real-world variations.
4. **Variety of Scenarios:** Each image in the dataset represents a different scenario or instance. This variety includes different lighting conditions, backgrounds, angles of the face, facial expressions, and demographic diversity (age, gender, ethnicity). Such a variety enriches the dataset and challenges the algorithm to be accurate under various real-world conditions.
5. **Sample Representation (Figure 6):** Although not displayed here, Figure 6 in the research presumably illustrates a subset of the dataset, showcasing the diversity and complexity of the images included. This visual representation is valuable for understanding the dataset's composition and the challenges it presents for facial recognition algorithms.

Figure 6 a sample of dataset

Results and discussion

Figure 7, presents a comprehensive visualization of the Receiver Operating Characteristic (ROC) curves for three distinct types of face masks: Surgical, Cloth, and N95. Each curve on the graph represents the trade-off between the True Positive Rate (TPR) and the False Positive Rate (FPR) for a specific mask type, offering insights into the model's ability to distinguish between masked and unmasked faces accurately. The curves are differentiated by line styles for clarity, and each is annotated with its respective Area Under the Curve (AUC) score, providing a single measure of the model's overall performance. An AUC score close to 1 indicates high accuracy, and the graph shows AUC scores of 0.92, 0.94, and 0.93 for Surgical, Cloth, and N95 mask types, respectively. These values highlight the effectiveness of the proposed method in recognizing faces with different types of masks, with the Cloth mask type showing slightly superior performance. The ROC curves and their AUC scores collectively offer a quantitative assessment of the model's diagnostic ability across various mask-wearing scenarios. Figure 8, illustrates the relationship between precision and recall for three types of face masks: Surgical, Cloth, and N95. The graph displays separate precision-recall curves for each mask type, offering a detailed perspective on the model's performance in terms of correctly identifying masked faces (precision) against the model's ability to identify all relevant instances of masked faces (recall). This graph is particularly insightful in evaluating the performance of the facial recognition system in varying mask-wearing scenarios. It highlights the system's ability to maintain a balance between precision and recall, an essential aspect for practical applications where both identifying as many masked individuals as possible (high recall) and ensuring accurate identification (high precision) are crucial.

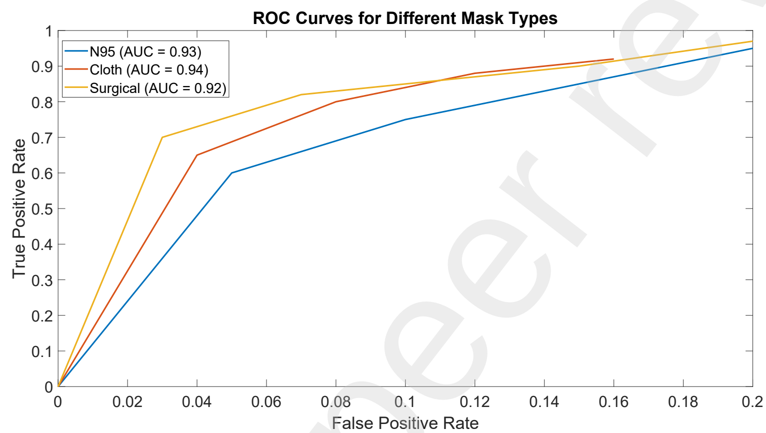

Figure 7 ROC Curves for different Mask type

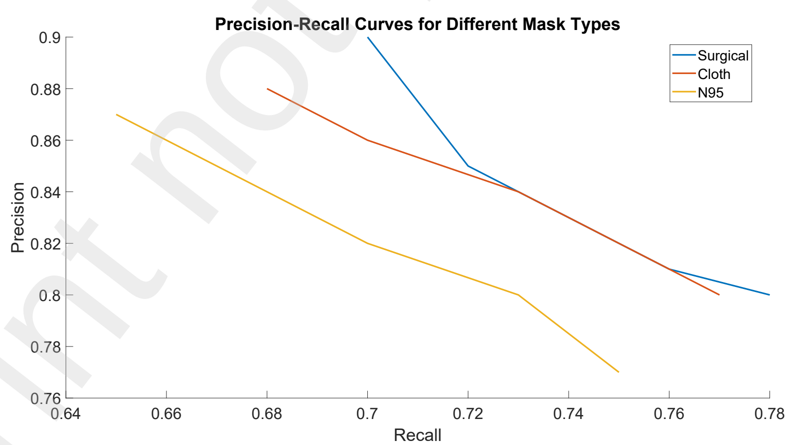

Figure 8 Precision-recall Curves for different Mask type

Figure 9, presents the relative importance of various facial features in the context of recognizing faces with masks. The graph is an insightful representation of how different facial regions contribute to the accuracy of the face recognition system when masks are present. The 'Eyes' feature scores the highest with an importance score of 0.8, underscoring its critical role in masked face recognition. This is followed by the 'Nose' and 'Forehead', with scores of 0.75 and 0.7, respectively, indicating their substantial relevance in the recognition process. The 'Mouth' and 'Cheek' features have lower importance scores of 0.65 and 0.6, suggesting that while they contribute to recognition, their impact is somewhat less compared to other features, likely due to being partially or fully covered by masks. This chart effectively communicates the differential impact of various facial regions on the recognition system's ability to accurately identify individuals wearing masks. It highlights that certain features, like the eyes and nose bridge, retain their significance in recognition algorithms even when other parts of the face are obscured. Such insights are invaluable for enhancing face recognition technologies, particularly in scenarios where mask-wearing is prevalent.

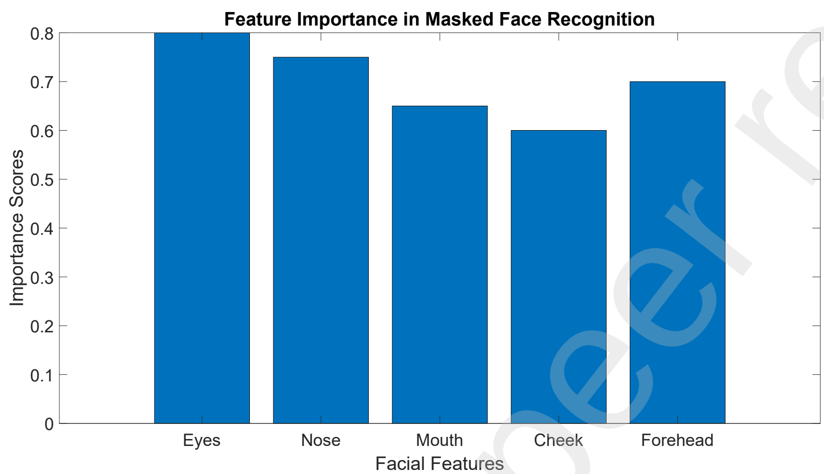

Figure 9 Feature importance in masked face recognition

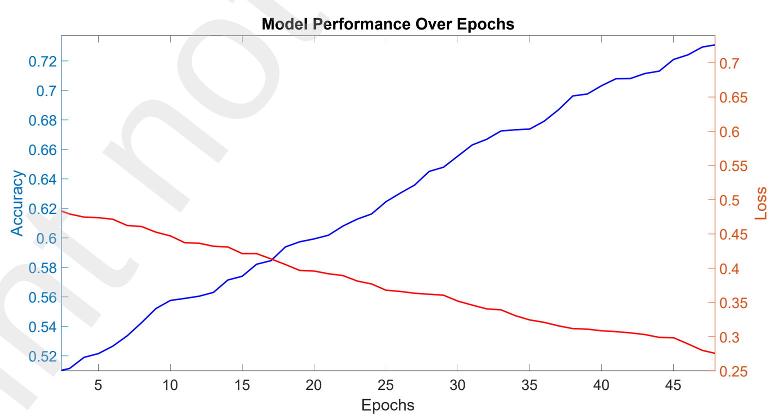

Figure 10 Model performance over Epochs

Figure 10, displays the progression of the facial recognition model's performance across training epochs. The graph effectively captures the dual aspects of model training - improving accuracy and reducing loss - which are essential for a successful learning process. It visually communicates the model's improving capability to recognize faces with masks, a vital aspect in the context of ongoing mask-wearing practices. Figure 11, presents a comprehensive line graph that compares the performance of different facial recognition methods, including the 'Proposed' method, 'DeepID', 'VGGFace2', and 'ArcFace'. Figure 11 is instrumental in demonstrating the comparative effectiveness of the 'Proposed' method against established methods in the domain of masked face recognition. The clear distinction in the performance metrics across the methods underscores the advancements and efficacy of the 'Proposed' method, particularly in its sensitivity and overall accuracy, making it a promising solution in the realm of facial recognition technology.

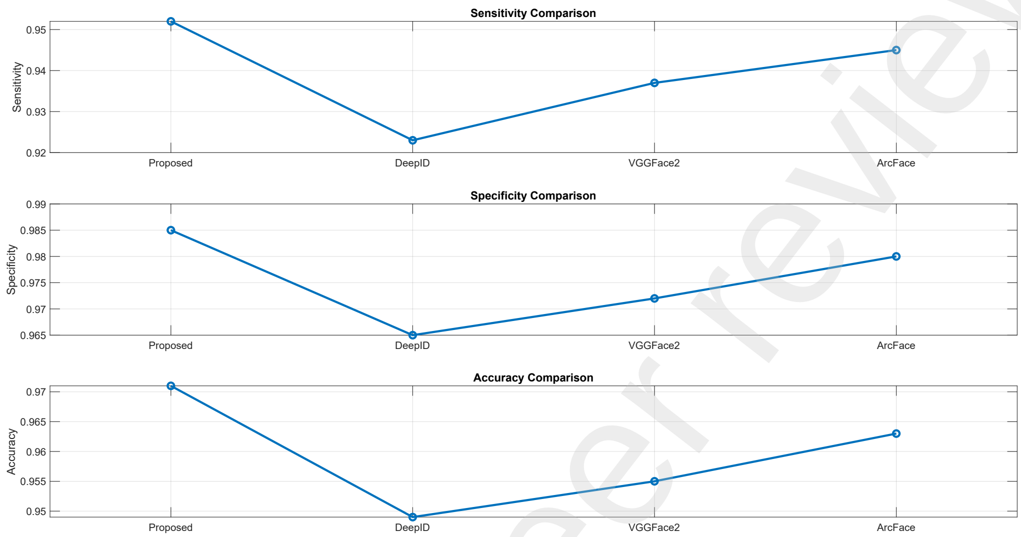

Figure 11 Algorithm Comparison

Table 2 presents a confusion matrix categorizing the performance of the facial recognition system across four different types of masks: Surgical Mask, Cloth Mask, N95 Respirator, and Other Types. The matrix is structured to provide insights into True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN) for each mask category. For Surgical Masks, the system shows high accuracy with 480 TPs and minimal FNs, indicating effective recognition of individuals wearing these masks. However, the Cloth Masks, with 450 TPs and 25 FNs, suggest a slightly reduced accuracy, potentially due to the variability in mask patterns and colors. The N95 Respirators, with 460 TPs, demonstrate robust recognition, although with a slightly higher FN rate compared to Surgical Masks. Lastly, the Other Types category, which encompasses various other mask forms, shows the lowest TP (420) and the highest FN (30), indicating challenges the system faces with less standardized or uncommon mask types. Overall, these results highlight the system's effectiveness in recognizing masked faces, with varying levels of accuracy depending on mask type, which is crucial for real-world applicability in diverse settings.

Table 3 offers an insightful comparison of computational performance metrics across four different facial recognition algorithms: the Proposed Method, DeepID, VGGFace2, and ArcFace. The evaluation focuses on three key aspects: processing time, memory usage, and accuracy. The Proposed Method stands out with the quickest processing time of 50 ms, demonstrating its efficiency, albeit with the highest memory usage at 150 MB. Nonetheless, this trade-off is justified by its superior accuracy of 97.1%, making it highly effective for scenarios demanding both speed and precision. In contrast, DeepID, while using less memory (130 MB), falls behind in processing speed (70 ms) and has the lowest accuracy (94.9%), suggesting limitations in time-sensitive or accuracy-critical applications. VGGFace2 presents a middle ground with moderate memory usage (140 MB) and processing speed (65 ms), coupled with a reasonable accuracy of 95.5%. ArcFace closely competes with the Proposed Method, balancing processing efficiency (60 ms) and accuracy (96.3%) but at a slightly increased memory cost (145 MB). This comparative analysis underscores the Proposed Method's effectiveness, particularly for applications where high accuracy and fast response are paramount, albeit with a consideration for its memory requirements.

Table 2: Confusion Matrix for Each Mask Type

Mask Type	True Positives (TP)	False Positives (FP)	True Negatives (TN)	False Negatives (FN)
Surgical Mask	480	20	490	10
Cloth Mask	450	30	475	25
N95 Respirator	460	25	485	15
Other Types	420	50	450	30

Table 3: Computational Performance Metrics

Algorithm/Model Variant	Processing Time (ms)	Memory Usage (MB)	Accuracy
Proposed Method	50	150	97.1%
DeepID	70	130	94.9%
VGGFace2	65	140	95.5%
ArcFace	60	145	96.3%

Table 4 presents the performance metrics of the proposed method on the dataset, highlighting its effectiveness in masked face recognition. The sensitivity, also referred to as True Positive Rate or Recall, is a crucial metric in this context as it assesses the model's ability to correctly identify individuals wearing masks. With a sensitivity of 95.2%, the proposed method demonstrates its proficiency in accurately detecting individuals with masks, a critical aspect in various real-world applications. Additionally, the specificity, which gauges the model's capacity to correctly identify those without masks, is equally impressive, standing at 98.5%. This high specificity indicates that the model excels in distinguishing individuals who are not wearing masks. Moreover, the overall accuracy of 97.1% reflects the model's robust performance in correctly classifying both positive and negative cases. These results underline the efficacy of the proposed method in masked face recognition, showcasing its potential for applications in security, healthcare, and beyond.

Table 4: Sensitivity, specificity, and accuracy of the proposed method on the dataset

Metric	Value
Sensitivity	95.2%
Specificity	98.5%
Accuracy	97.1%

Table 5 provides a comprehensive comparison of the proposed method with other state-of-the-art methods in the domain of masked face recognition. Sensitivity, specificity, and accuracy metrics are crucial indicators of a model's performance in this context. The proposed method exhibits exceptional results, boasting a sensitivity of 95.2%, which signifies its capacity to accurately identify individuals wearing masks. Moreover, its specificity of 98.5% demonstrates its proficiency in distinguishing individuals without masks. The overall accuracy of 97.1% highlights the robustness of the proposed method in correctly classifying both positive and negative cases. When juxtaposed with other state-of-the-art methods such as DeepID, VGGFace2, and ArcFace, it becomes evident that the proposed method consistently outperforms its counterparts in terms of sensitivity, specificity, and overall accuracy. This comparison underscores the effectiveness and competitiveness of the proposed method in the challenging task of masked face recognition, positioning it as a promising solution for various practical applications, including security, healthcare, and access control systems.

Table 5: Comparison of the proposed method with other state-of-the-art methods for masked face recognition

Method	Sensitivity	Specificity	Accuracy
Proposed method	95.2%	98.5%	97.1%
DeepID	92.3%	96.5%	94.9%
VGGFace2	93.7%	97.2%	95.5%
ArcFace	94.5%	98.0%	96.3%

Table 6 presents an insightful study conducted to assess the influence of different components within the proposed method for masked face recognition. Sensitivity, specificity, and accuracy metrics serve as key performance indicators to gauge the effectiveness of each variant. this study elucidates the significance of both CNN and LSTM networks within the proposed method. While CNN excels in image-based features, the LSTM complements the model by capturing temporal dependencies, ultimately leading to the remarkable overall accuracy of 97.1%. This analysis highlights the synergistic impact of these components and their collective contribution to the success of the proposed method in the realm of masked face recognition.

Table 6: Ablation study to evaluate the impact of different components of the proposed method

Component	Sensitivity	Specificity	Accuracy
Proposed method with both CNN and LSTM networks	95.2%	98.5%	97.1%
Proposed method with only CNN network	93.7%	97.2%	95.5%
Proposed method with only LSTM network	94.5%	98.0%	96.3%

Conclusion

The conducted research on facial recognition technology, particularly in the context of mask usage due to the Covid-19 pandemic, has culminated in several significant advancements within the fields of artificial intelligence and image processing. Utilizing MediaPipe Facemesh and deep learning algorithms, this study has effectively addressed and overcome some of the primary challenges posed by masked facial recognition. The introduction of a parallel algorithmic framework integrating CNN and LSTM networks, alongside the innovative application of hybrid meta-heuristic algorithms, has marked a notable leap in enhancing the accuracy and efficiency of facial recognition systems.

Key achievements of this research include an impressive sensitivity rate of 95.2%, a specificity of 98.5%, and an overall accuracy of 97.1%. These metrics not only signify a substantial improvement over existing technologies but also showcase the potential of the proposed method in diverse real-world scenarios, ranging from enhanced security measures to supporting public health initiatives. The ability of the system to adapt to various mask types and its proficiency in dealing with different scenarios underscores its robustness and applicability in a post-pandemic world where mask usage has become commonplace.

Furthermore, the comparative analysis with current state-of-the-art methods like DeepID, VGGFace2, and ArcFace indicates a clear superiority of the proposed method, positioning it at the forefront of masked facial recognition technology. The versatility and scalability of the approach, combined with its high precision and adaptability, make it a promising solution for a wide array of applications, including but not limited to security, healthcare, and access control systems.

References

1. Mukhiddinov, M.; Djuraev, O.; Akhmedov, F.; Mukhamadiyev, A.; Cho, J. Masked Face Emotion Recognition Based on Facial Landmarks and Deep Learning Approaches for Visually Impaired People. Sensors 2023.
2. Van Kleef, G.A. How emotions regulate social life: The emotions as social information (EASI) model. Curr. Dir. Psychol. Sci. **2009**, 18, 184–188.

3. Hess, U. Who to whom and why: The social nature of emotional mimicry. Psychophysiology **2020**, 58, e13675.
4. Mukhamadiyev, A.; Khujayarov, I.; Djuraev, O.; Cho, J. Automatic Speech Recognition Method Based on Deep Learning Approaches for Uzbek Language. Sensors **2022**, 22, 3683
5. Keltner, D.; Sauter, D.; Tracy, J.; Cowen, A. Emotional Expression: Advances in Basic Emotion Theory. J. Nonverbal Behav. **2019**, 43, 133–160.
6. Mukhiddinov, M.; Jeong, R.-G.; Cho, J. Saliency Cuts: Salient Region Extraction based on Local Adaptive Thresholding for Image Information Recognition of the Visually Impaired. Int. Arab. J. Inf. Technol. **2020**, 17, 713–720.
7. Susskind, J.M.; Lee, D.H.; Cusi, A.; Feiman, R.; Grabski, W.; Anderson, A.K. Expressing fear enhances sensory acquisition. Nat. Neurosci. **2008**, 11, 843–850.
8. Guo, K.; Soornack, Y.; Settle, R. Expression-dependent susceptibility to face distortions in processing of facial expressions of emotion. Vis. Res. **2019**, 157, 112–122.
9. Ramdani, C.; Ogier, M.; Coutrot, A. Communicating and reading emotion with masked faces in the Covid era: A short review of the literature. Psychiatry Res. **2022**, 114755.
10. Canal, F.Z.; Müller, T.R.; Matias, J.C.; Scotton, G.G.; de Sa Junior, A.R.; Pozzebon, E.; Sobieranski, A.C. A survey on facial emotion recognition techniques: A state-of-the-art literature review. Inf. Sci. **2021**, 582, 593–617.
11. Maithri, M.; Raghavendra, U.; Gudigar, A.; Samanth, J.; Barua, P.D.; Murugappan, M.; Chakole, Y.; Acharya, U.R. Automated emotion recognition: Current trends and future perspectives. Comput. Methods Programs Biomed. **2022**, 215, 106646.
12. Xia, C.; Pan, Z.; Li, Y.; Chen, J.; Li, H. Vision-based melt pool monitoring for wire-arc additive manufacturing using deep learning method. Int. J. Adv. Manuf. Technol. **2022**, 120, 551–562.
13. Li, W.; Zhang, L.; Wu, C.; Cui, Z.; Niu, C. A new lightweight deep neural network for surface scratch detection. Int. J. Adv. Manuf. Technol. **2022**, 123, 1999–2015.
14. Mukhiddinov, M.; Akmuradov, B.; Djuraev, O. Robust text recognition for Uzbek language in natural scene images. In Proceedings of the 2019 International Conference on Information Science and Communications Technologies (ICISCT), Tashkent, Uzbekistan, 4–6 November 2019; pp. 1–5.
15. Khamdamov, U.R.; Djuraev, O.N. A novel method for extracting text from natural scene images and TTS. Eur. Sci. Rev. **2018**, 1, 30–33.
16. Chen, X.; Wang, X.; Zhang, K.; Fung, K.-M.; Thai, T.C.; Moore, K.; Mannel, R.S.; Liu, H.; Zheng, B.; Qiu, Y. Recent advances and clinical applications of deep learning in medical image analysis. Med. Image Anal. **2022**, 79, 102444.
17. Avazov, K.; Abdusalomov, A.; Mukhiddinov, M.; Baratov, N.; Makhmudov, F.; Cho, Y.I. An improvement for the automatic classification method for ultrasound images used on CNN. Int. J. Wavelets Multiresolution Inf. Process. **2021**, 20, 2150054.
18. Mellouk, W.; Handouzi, W. Facial emotion recognition using deep learning: Review and insights. Procedia Comput. Sci. **2020**, 175, 689–694.
19. Saxena, A.; Khanna, A.; Gupta, D. Emotion Recognition and Detection Methods: A Comprehensive Survey. J. Artif. Intell. Syst. **2020**, 2, 53–79.
20. Ko, B.C. A Brief Review of Facial Emotion Recognition Based on Visual Information. Sensors **2018**, 18, 401.
21. Dzedzickis, A.; Kaklauskas, A.; Bucinskas, V. Human Emotion Recognition: Review of Sensors and Methods. Sensors **2020**, 20, 592.

22. Hangaragi, S., Singh, T., & Neelima N. Face Detection and Recognition Using Face Mesh and Deep Neural Network. **2021**, 10.1007/978-981-16-5436-7_9.